

1.a

Recall.

The received signal at D_n can be expressed as

$$y_{Dn} = h_{SDn} y_s + h_{fn} y_{uf} + h_{nn} y_{un} + \overbrace{n_n}^{\text{noise}}$$

$$y_s = \sqrt{a_f P_s} x_f + \sqrt{a_n P_s} x_n, \quad \begin{matrix} y_{uf} = \sqrt{P_{uf}} s_f \\ y_{un} = \sqrt{P_{un}} s_n \end{matrix} \quad \begin{matrix} r_n \sim \mathcal{CN}(0, N_0) \\ E[s_n^2] = 1 \end{matrix}$$

$a_f + a_n = 1$ $a_f > a_n$; $E[s_n^2] = 1$ $E[s_f^2] = 1$

According to NOMA principle : for D_n x_f is first decoded and then decode x_n after eliminating x_f by SIC

$\therefore$ the receive signal-to-interference-plus-noise ratio (SINR) of x_f and x_n at D_n are respectively given by

$$\gamma_{Dn}^{x_f} = \frac{a_f \frac{P_s}{N_0} |h_{SDn}|^2}{a_n \frac{P_s}{N_0} |h_{SDn}|^2 + \frac{P_{uf}}{N_0} |h_{fn}|^2 + \frac{P_{un}}{N_0} |h_{nn}|^2 + 1}$$

$$\gamma_{Dn}^{x_n} = \frac{a_n \frac{P_s}{N_0} |h_{SDn}|^2}{\frac{P_{uf}}{N_0} |h_{fn}|^2 + \frac{P_{un}}{N_0} |h_{nn}|^2 + 1}$$

The outage event occurs at D_n , when either x_f or x_n cannot be decoded at D_n

The outage probability of D_n can be expressed as

$$P_{out}^{D_n} = 1 - \Pr \left(\underbrace{\gamma_{Dn}^{x_f} > \gamma_{thf}^{DL}}_{\text{term 1}}, \underbrace{\gamma_{Dn}^{x_n} > \gamma_{thn}^{DL}}_{\text{term 2}} \right)$$

γ_{thn}^{DL} , γ_{thf}^{DL} are SNR threshold of x_n and x_f respectively.

Consider $\gamma_{Dn}^{x_f} > \gamma_{thf}^{DL}$
term 1

$$= \frac{a_f \frac{P_s}{N_0} |h_{SDn}|^{\alpha}}{a_n \frac{P_s}{N_0} |h_{SDn}|^{\alpha} + \frac{P_{uf}}{N_0} |h_{fu}|^{\alpha} + \frac{P_{un}}{N_0} |h_{nn}|^{\alpha} + 1} > \gamma_{thf}^{DL}$$

$$= a_f \frac{P_s}{N_0} |h_{SDn}|^{\alpha} > \left(a_n \frac{P_s}{N_0} |h_{SDn}|^{\alpha} + \frac{P_{uf}}{N_0} |h_{fu}|^{\alpha} + \frac{P_{un}}{N_0} |h_{nn}|^{\alpha} + 1 \right) \gamma_{thf}^{DL}$$

$$= (a_f - a_n \gamma_{thf}^{DL}) \frac{P_s}{N_0} |h_{SDn}|^{\alpha} > \left(\frac{P_{uf}}{N_0} |h_{fu}|^{\alpha} + \frac{P_{un}}{N_0} |h_{nn}|^{\alpha} + 1 \right) \gamma_{thf}^{DL}$$

$$|h_{SDn}|^{\alpha} > \gamma_{thf}^{DL} \frac{\left(\frac{P_{uf}}{N_0} |h_{fu}|^{\alpha} + \frac{P_{un}}{N_0} |h_{nn}|^{\alpha} + 1 \right)}{(a_f - a_n \gamma_{thf}^{DL}) \frac{P_s}{N_0}}$$

$$\text{let } \gamma_s = \frac{P_s}{N_0}, \quad \gamma_{uf} = \frac{P_{uf}}{N_0}, \quad \gamma_{un} = \frac{P_{un}}{N_0}$$

$$\theta_f = \frac{\gamma_{thf}^{DL}}{(a_f - a_n \gamma_{thf}^{DL}) \frac{P_s}{N_0}}$$

$$= |h_{SDn}|^{\gamma} > \theta_f (\gamma_{uf} |h_{fn}|^{\gamma} + \gamma_{un} |h_{nn}|^{\gamma} + 1)$$

Ug consider term 2

$$\gamma_{Dn}^{an} > \gamma_{thn}^{DL}$$

$$= \frac{a_n \frac{P_s}{N_0} |h_{SDn}|^{\gamma}}{\frac{P_{uf}}{N_0} |h_{fn}|^{\gamma} + \frac{P_{un}}{N_0} |h_{nn}|^{\gamma} + 1} > \gamma_{thn}^{DL}$$

$$= |h_{SDn}|^{\gamma} > \frac{\gamma_{thn}^{DL}}{\gamma_s a_n} (\gamma_{uf} |h_{fn}|^{\gamma} + \gamma_{un} |h_{nn}|^{\gamma} + 1)$$

$$\text{let } \theta_n = \frac{\gamma_{thn}^{DL}}{a_n \gamma_s}$$

$$|h_{SDn}|^{\gamma} > \theta_n (\gamma_{uf} |h_{fn}|^{\gamma} + \gamma_{un} |h_{nn}|^{\gamma} + 1)$$

w.k.t

$|h_i| \rightarrow$ follows Rayleigh distribution

$|h_i|^{\gamma} -$ follows Exponential distribution

PDF of $|h_i|^2$ is

$$f_{|h_i|^2}(x) = \frac{1}{\beta_i} e^{-x/\beta_i} \quad E[|h_i|^2] = \beta_i$$

CDF of $|h_i|^2$ is

$$F_{|h_i|^2}(x) = 1 - e^{-x/\beta_i}$$

RVS

$$\text{Let } Z = |h_{SDN}|^2, \quad Y = |h_{fn}|^2, \quad X = |h_{nn}|^2$$

$$\text{and mean, } E[Z] = \beta_{SDN}, \quad E[Y] = \beta_{fn}, \quad E[X] = \beta_{nn}$$

$$P_{out}^{Dn} = 1 - \Pr \left[|h_{SDN}|^2 > \theta_f (\gamma_{uf} |h_{fn}|^2 + \gamma_{un} |h_{nn}|^2 + 1), \right. \\ \left. |h_{SDN}|^2 > \theta_n (\gamma_{uf} |h_{fn}|^2 + \gamma_{un} |h_{nn}|^2 + 1) \right]$$

$\therefore$

$$P_{out}^{Dn} = 1 - \Pr \left[Z > \theta_f (\gamma_{uf} Y + \gamma_{un} X + 1), Z > \theta_n (\gamma_{uf} Y + \gamma_{un} X + 1) \right]$$

$$= 1 - \Pr \left[Z > \max(\theta_f, \theta_n) (\gamma_{uf} Y + \gamma_{un} X + 1) \right]$$

$$\text{let } \theta = \max(\theta_f, \theta_n)$$

$$1 - \Pr \left[Z > \theta (\gamma_{uf} Y + \gamma_{un} X + 1) \right]$$

consider $\Pr [Z > 0 \text{ (} \gamma_{uf} Y + \gamma_{un} X + 1 \text{)}]$

$$= \int_0^{\infty} f_Z(z) \int_0^{\infty} f_Y(y) \int_0^{\infty} f_X(x) \mathbb{1}_{\theta(\gamma_{uf} y + \gamma_{un} x + 1)} dz dy dx$$

$$= \int_0^{\infty} \frac{1}{\beta_{SDn}} e^{-z/\beta_{SDn}} dz \int_0^{\infty} \frac{1}{\beta_{fn}} e^{-y/\beta_{fn}} dy \int_0^{\infty} \frac{1}{\beta_{nn}} e^{-x/\beta_{nn}} dx$$

$$= \frac{e^{-\theta(\gamma_{uf} y + \gamma_{un} x + 1)/\beta_{SDn}}}{e^{\frac{\beta_{SDn}}{\beta_{fn}}}} \int_0^{\infty} \frac{1}{\beta_{fn}} e^{-y/\beta_{fn}} dy \int_0^{\infty} \frac{1}{\beta_{nn}} e^{-x/\beta_{nn}} dx$$

$$= e^{-\theta(\gamma_{un} x + 1)/\beta_{SDn}} \int_0^{\infty} \frac{1}{\beta_{fn}} e^{-\left(\frac{\theta \gamma_{uf}}{\beta_{SDn}} + \frac{1}{\beta_{fn}}\right)y} dy \int_0^{\infty} \frac{1}{\beta_{nn}} e^{-x/\beta_{nn}} dx$$

$$= e^{-\frac{\theta(\gamma_{un} x + 1)}{\beta_{SDn}}} \left(\frac{\beta_{SDn}}{\theta \gamma_{uf} \beta_{fn} + \beta_{SDn}} \right) \int_0^{\infty} \frac{1}{\beta_{nn}} e^{-x/\beta_{nn}} dx$$

$$= e^{-\theta/\beta_{SDn}} \left(\frac{\beta_{SDn}}{\theta \gamma_{uf} \beta_{fn} + \beta_{SDn}} \right) \int_0^{\infty} \frac{1}{\beta_{nn}} e^{-\left(\frac{\theta \gamma_{un}}{\beta_{SDn}} + \frac{1}{\beta_{nn}}\right)x} dx$$

$$\left(\frac{\beta_{SDn}}{\theta \gamma_{uf} \beta_{fn} + \beta_{SDn}} \right) e^{-\theta/\beta_{SDn}} \left(\frac{\beta_{SDn}}{\theta \gamma_{un} \beta_{nn} + \beta_{SDn}} \right)$$

$$= \frac{(\beta_{SDn})^2 \cdot e^{-\theta/\beta_{SDn}}}{(\theta_{out} \beta_{fn} + \beta_{SDn})(\theta_{un} \beta_{nn} + \beta_{SDn})}$$

∴ The outage probability

$$P_{out}^{DL} = 1 - \frac{(\beta_{SDn})^2 \cdot e^{-\theta/\beta_{SDn}}}{(\theta_{out} \beta_{fn} + \beta_{SDn})(\theta_{un} \beta_{nn} + \beta_{SDn})}$$

Hence proved.